Question	Answer	Marks	Partial Marks
1(a)(i)	11.61 final answer	**2**	**M1** for $13.5[0] \times \left(1 - \frac{14}{100}\right)$ oe or **B1** for 1.89
1(a)(ii)	197.37 final answer	**2**	**FT** 17 × *their* (a)(i) exact or correct to nearest cent **M1** for 42.5 ÷ 2.5
1(b)(i)	53.3 or 53.33…	**1**	
1(b)(ii)	7.5	**2**	**M1** for 22.5 ÷ (2 + 8 + 5) oe soi
1(c)	20.55×2.45 oe	**M2**	**M1** for 20.5 + 0.05 oe seen or 2.4 + 0.05 oe seen If 0 scored, **SC1** here for 20.45×2.35 oe
	3 nfww	**A2**	**M1** for *their* area ÷ 10 ÷ 2.5 oe
2(a)(i)	1, –0.5 oe	**2**	**B1** for each
2(a)(ii)	Correct curve	**4**	**B3FT** for 6 or 7 correct plots or **B2FT** for 4 or 5 correct plots or **B1FT** for 2 or 3 correct plots
2(b)	$y = 2.5 - 2x$ ruled	**B2**	**B1** for $y = k - 2x$ or $y = px + 2.5$ ruled ($p \neq 0$) or for [y =] $2.5 - 2x$ oe identified
	1.3 to 1.4	**B1**	
2(c)	–1	**B1**	
	$y = -1$	**B1**	**FT** *their k* (must be negative)
3(a)(i)	7^{11} cao	**1**	
3(a)(ii)	7^{10} cao	**1**	
3(a)(iii)	7^2 cao	**1**	If answers 11, 10 and 2 in **(a)** then allow **SC1** in this part
3(b)	$1000x^9y^{12}$ final answer	**3**	**B2** for correct answer seen or answer of the form $1000x^9y^k$ or $1000x^ky^{12}$ or kx^9y^{12} or **B1** for answer with one correct element in product or $(10x^3y^4)^{[3]}$ seen
3(c)(i)	108	**2**	**M1** for [540 =] 2^2 [×] 3^3 [×] 5 or **B1** for 108 oe not in prime factor form e.g. $2^2 \times 3 \times 9$

Question	Answer	Marks	Partial Marks
3(c)(ii)	30 240	**2**	**M1** for $(540 \times 2^5 \times 3^3 \times 7) \div$ *their* **(c)(i)** oe or **B1** for answer 30 240 oe not in prime factor form e.g. $2^5 \times 3^3 \times 35$
3(c)(iii)	98	**2**	**B1** for 592 704 seen or $2^6 \times 3^3 \times 7^3$ seen or 2×7^2 oe seen
3(d)(i)	$(x - 7)(x + 4)$ final answer	**2**	**M1** for $x(x - 7) + 4(x - 7)$ or $x(x + 4) - 7(x + 4)$ or better or for $(x + a)(x + b)$ where $ab = -28$ or $a + b = -3$
3(d)(ii)	$(a + 2b)(11a + 14b)$ final answer	**2**	**M1** for $(a + 2b)(7(a + 2b) + 4a)$ or $(a + pb)(11a + qb)$ where $pq = 28$ or $11p + q = 36$ If 0 scored, **SC1** for $a + 2b\ (11a + 14b)$
3(e)	$[y =]\dfrac{5x-1}{2}$ oe final answer	**4**	**B2** for $2x - 1 = -2x + 2y - x$ oe or **B1** for $9^x = 3^{2x}$ or better **M1dep** for correct rearrangement of *their* 5 term 'linear' equation in y and x to make y the subject
4(a)(i)	Correct histogram	**3**	**B1** for each correct block If 0 scored**, SC1** for any two of fds 7.5, 3.33…, 0.8 oe soi
4(a)(ii)	3.7875 or 3.79 or 3.787 or 3.788	**4**	**M1** for 0.75, 1.5, 3, 5.5, 9.5 soi **M1** for Σfx **M1 dep** for *their* $\Sigma fx \div 40$
4(a)(iii)	$\dfrac{11}{40}$ oe	**1**	
4(a)(iv)	$\dfrac{30}{203}$ oe	**3**	**M2** for $[2 \times]\dfrac{4}{29} \times \dfrac{15}{28}$ oe or **M1** for $\dfrac{4}{29}$ or $\dfrac{15}{29}$ oe seen After 0 scored, **SC1** for $[2\times]\left(\dfrac{4}{40} \times \dfrac{26}{39}\right)$ oe or for answer $\dfrac{120}{841}$ oe
4(b)(i)	4.6	**1**	
4(b)(ii)	3.2	**1**	